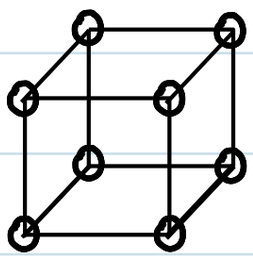
31 表.

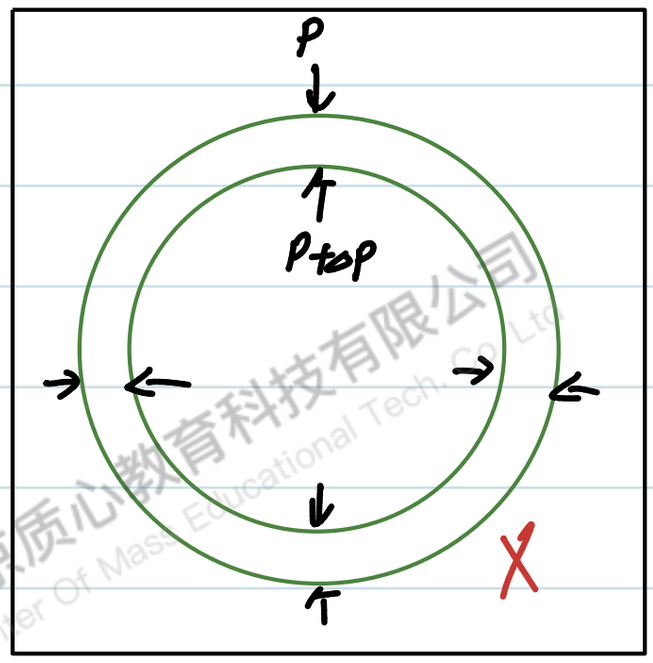
1. 中子间距 r_0.

$r_0^3 \cdot n = 1$



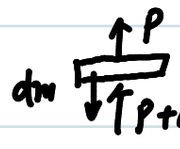
$8 \cdot \frac{1}{8} = 1$





$$F_{31} = \frac{G \cdot \frac{4}{3}\pi r^3 \cdot n \cdot m_0 \cdot m}{r^2}$$

$$= \frac{4}{3}\pi G n m_0 \cdot m \cdot r.$$



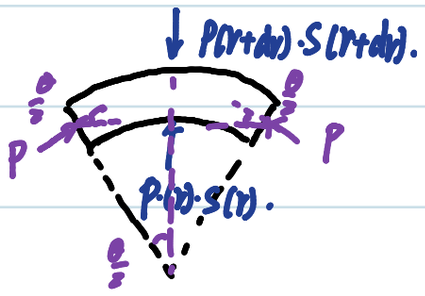
$$dp \cdot S = \frac{4}{3}\pi G n m_0 r \cdot S\, dr \cdot n m_0$$

$$\Rightarrow dp = \frac{4}{3}\pi G n^2 m_0^2 r \cdot dr.$$

$$\Rightarrow p_0 = \frac{2}{3}\pi G n^2 m_0^2 R^2.$$

由题. $p_0 = A \frac{n_0}{m_0}\left(\frac{h}{r_0}\right)^2$.

$$\Rightarrow R = \sqrt{\frac{3Ah^2}{2\pi G m_0^2 n^{\frac{1}{3}}}}$$

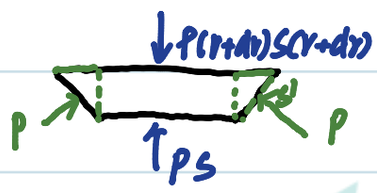


场. 可以用算.

$$\nabla \cdot \vec{g} = \frac{\partial g_r}{\partial r} + \frac{g_r}{r}.$$

$\nabla \cdot \vec{F}$ = 直. $\frac{\partial F_x}{\partial x}$

= 球. $\frac{\partial F_r}{\partial r} + \frac{2F_r}{r}$.



$2P \cdot S' \cdot \cos\theta$

$= P \cdot (2S' \cdot \cos\theta) = P \cdot \Delta S$.

场全布 $\longleftrightarrow$ 力密度全布.

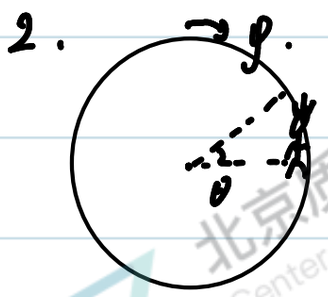
势全布 $\longleftrightarrow$ 压强全布.

$\Rightarrow$. 静力学. 侧压力. 与顶压强多出来的部分刚好抵掉.

算太阳风 压强. 直 v.

热流. 辐流. 能流. 人流 直 x 球 v.

2.



(1). 角: $mr^2\theta - \frac{1}{2}mr^2\varphi = 0$

约: $\theta + \varphi = 2\pi$

$\Rightarrow \varphi = \frac{2}{3} \cdot 2\pi$.

(2).



$\rho = \frac{r}{2} \cdot \frac{1}{\cos\theta}$

$I = m\rho^2$.

角: $m\rho^2 d\theta - \frac{1}{2}mr^2 \cdot d\varphi = 0$.

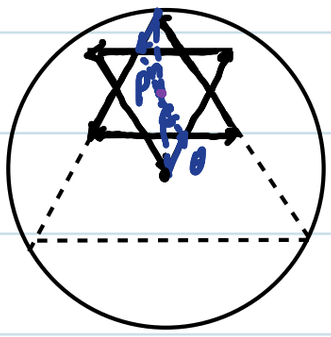
$\Rightarrow$. $d\varphi = 2\frac{\rho^2}{r^2}d\theta$.

n: $\frac{1}{2}\rho^2 d\theta = ds$.

$\Rightarrow$. $d\varphi = 4\frac{1}{\pi^2}ds$.

$\Rightarrow$ $\varphi = \frac{4S}{r^2}. = \frac{4 \times \frac{\sqrt{3}}{4}(\sqrt{3}r)^2}{r^2} = 3\sqrt{3}$

(3).



先想 人→地 三角形. 边长 $\frac{\sqrt{3}}{2}r$.

角: $m\rho^2 d\theta + m\rho^2 d\theta - \frac{1}{2}mr^2 d\varphi = 0$.

$\Rightarrow$ $2\rho^2 d\theta = \frac{1}{2}r^2 d\varphi$.

$\Rightarrow$ $4S = \frac{1}{2}r^2 \cdot \varphi$.

$\Rightarrow$ $4 \cdot \frac{\sqrt{3}}{4}\left(\frac{\sqrt{3}}{2}r\right)^2 = \frac{1}{2}r^2 \cdot \varphi$.

$\Rightarrow$. $\varphi = \frac{3\sqrt{3}}{2}$

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3. $\Delta P = \frac{f}{S} = \frac{P_0}{4}$

(1). 升温至 $2T_0$ 有 V_{max}.

$(P_0+\Delta P)\cdot V_{max} = \frac{P_0 V_0}{T_0}\cdot 2T_0$

$\Rightarrow V_{max} = \frac{8}{5}V_0$

降温至 $\frac{1}{5}T_0$ 有 V_{min}

$(P_0-\Delta P)\cdot V_{min} = \frac{P_0 V_0}{T_0}\cdot \frac{1}{5}T_0$

$\Rightarrow V_{min} = \frac{2}{3}V_0$.

(2). $W = (P_0+\Delta P)(V_{max}-V_{min})$.

$-(P_0-\Delta P)(V_{max}-V_{min})$

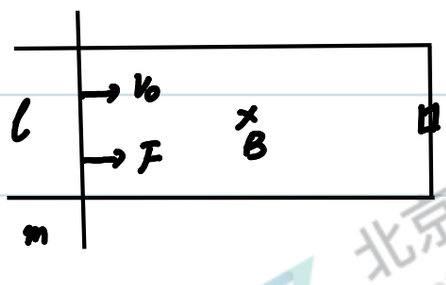
$= \frac{7}{15}P_0V_0$.

(3). $Q = n C_V \Delta T + W'$

$= \frac{59}{15}P_0V_0$.

$\eta = \frac{W}{Q} = \frac{28}{295}$

4.



(1). $2ax_1 = v_0^2 \Rightarrow x_1 = 1m$.

(2) $\mathcal{E} = BVl$

$F = -BIl$.

$\Rightarrow F = -\frac{B^2l^2V}{R}$.

$F - \frac{1}{2}\cdot\frac{B^2l^2v_0}{R} = -ma$

$\Rightarrow F_1 = ?$

or.

$F - \frac{1}{2}\frac{B^2l^2(-v_0)}{R} = -ma$.

$\Rightarrow F_2 = ?$

(3). $V = v_0 - at$.

$F - \frac{B^2l^2V}{R} = -ma$.

$W = \frac{1}{2}mv^2 - \frac{1}{2}mv_0^2 + Q$.

$dQ = (\frac{B(v_0-at)}{R})^2 R\cdot dt$.

$= \frac{B^2l^2}{R}(v_0-at)^2 dt$.

$= \frac{B^2l^2}{R}(v_0^2\cdot t - v_0 a t^2 + \frac{1}{3}a^2t^3)$.

$\Rightarrow W = \frac{1}{2}m\cdot(v_0-at)^2 - \frac{1}{2}mv_0^2\cdot$

$+ \frac{B^2l^2}{R}(v_0^2\cdot t - v_0 a t^2 + \frac{1}{3}a^2t^3)$.

5. (1).

$-\frac{dB}{dt}\pi r^2\cdot\frac{1}{2\pi r}\cdot q = ma_\tau = m\frac{dv}{dt}$.

$\Rightarrow dv = -\frac{qr}{2m}dB$.

$\Rightarrow v_0 = v_0 - \frac{qr}{2m}B_0$

(2). $B_0 \mapsto -B_0$.

$v_0 \longmapsto v = v_0 - \frac{qr}{2m}(-2B_0)$.

$= v_0 + \frac{qr}{m}B_0$.

洛仑兹力不做功.

$W = \frac{1}{2}m(v_0+\frac{qr}{m}B_0)^2 - \frac{1}{2}mv_0^2$.

6. 自行推一下多普勒.

(1). $f = \frac{c}{c-v}f_0$.

(2) $f = \frac{c}{c-v\cos\varphi}\cdot f_0$.

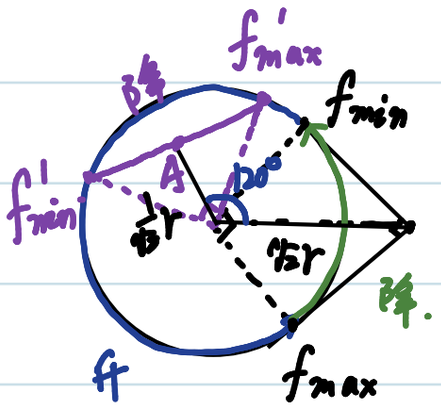
(3). 读图: $T = 24s$.

频率升高. $18s = t_1$ 频率下降 $6s = t_2$ $\frac{6}{24} = \frac{1}{4}$

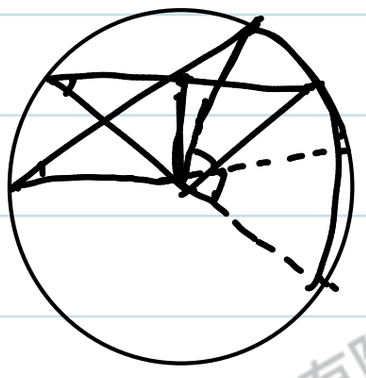
极值错开. $8s = \Delta t$. 极值数不一样. $\frac{8}{24} = \frac{1}{3}$.

$\left.\begin{array}{l}\frac{t_1}{T} = \frac{t_2}{T}\\ \text{极值不一样}\end{array}\right\} \Rightarrow$ 一内一外.

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B.



曲线1.

$$f_{max} = \frac{c}{c-v} f_0$$

$$f_{min} = \frac{c}{c+v} f_0.$$

$$\Rightarrow V = \frac{f_{max} - f_{min}}{f_{max} + f_{min}} \cdot c = 66 m/s.$$

$$\Rightarrow 2\pi r = V \cdot T.$$

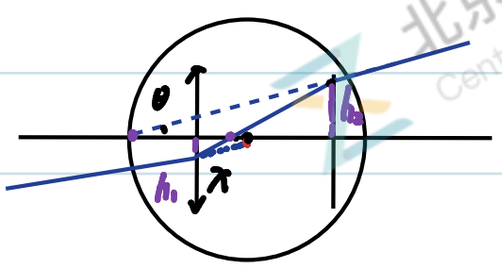
$$\Rightarrow r = \frac{V \cdot T}{2\pi} = 252 m.$$

$$\overline{AB}^2 = (\frac{1}{\sqrt{3}} r)^2 + (\frac{1}{2} r)^2 - 2 \cdot \frac{1}{\sqrt{3}} r \cdot \frac{1}{2} r \cos 120°$$

$$= \frac{7}{12} r^2.$$

$$\Rightarrow \overline{AB} = \sqrt{\frac{7}{3}} \cdot r = 411 m.$$

7. 共轭.



第1次成像. $S_1 = -xS$

$$\frac{1}{S_1} + \frac{1}{-xS} = \frac{1}{f}$$

$$\Rightarrow S_1' = \frac{f \cdot xS}{f + xS}$$

第2次成像. $S_2 = d - S_1'$

$$\frac{1}{S_2} + \frac{1}{S_2'} = \frac{1}{f}.$$

几何: $\theta \cdot (-S_1) = h_1$

$$\theta \cdot (-S_2) = h_2.$$

$$\frac{h_2}{h_1} = \frac{d - S_1'}{S_1'}$$

$$\Rightarrow xS = \frac{f \cdot d}{2f - d}.$$

$x + f = S.$

$\Rightarrow x = S - f$

8. 火箭相对地面参考系速度为 v 时.

在火箭参考系. 经时间. 速度增加 dv.

$$da = a\,dt.$$

$$v + dv = \frac{dv + v}{1 + \frac{v \cdot dv}{c^2}} \Rightarrow dv = \frac{1}{\gamma^2} da.$$

$$\Rightarrow a\,dt = da = \gamma^2 dv = \frac{dv}{1 - \beta^2}$$

$$\Rightarrow 2a\,dt = \frac{dv}{1-\beta} + \frac{dv}{1+\beta}$$

$$\Rightarrow \frac{2a}{c} dt = \ln \frac{c+v}{c-v}.$$

$$\Rightarrow \frac{c+v}{c-v} = e^{\frac{2at}{c}}.$$

$$\Rightarrow v(t) = \frac{e^{\frac{at}{c}} - e^{-\frac{at}{c}}}{e^{\frac{at}{c}} + e^{-\frac{at}{c}}} \cdot c.$$

$$\Rightarrow \gamma(t) = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}} = \frac{e^{\frac{at}{c}} + e^{-\frac{at}{c}}}{2}$$

设返上时间. 火箭参照系时间为 t_0.

$$\int_0^{t_0} v(t) \cdot \gamma(t) \cdot dt = \int_0^{t_0} v_0 \cdot \gamma(t) \cdot dt.$$

$$\Rightarrow \int_0^{t_0} \frac{e^{\frac{at}{c}} - e^{-\frac{at}{c}}}{2} c \cdot dt = \int_0^{t_0} v_0 \frac{e^{\frac{at}{c}} + e^{-\frac{at}{c}}}{2} dt.$$

$$\Rightarrow \frac{c}{a} c \left(e^{\frac{at_0}{c}} - 1\right) + \frac{c}{a} \cdot c \left(e^{-\frac{at_0}{c}} - 1\right) = \frac{c}{a} v_0 \left(e^{\frac{at_0}{c}} - 1\right) - \frac{c}{a} v_0 \left(e^{-\frac{at_0}{c}} - 1\right).$$

$$\Rightarrow (c - v_0) e^{\frac{at_0}{c}} + (c + v_0) e^{-\frac{at_0}{c}} = 2c.$$

$\beta_0 = \frac{v_0}{c}$. $k = \frac{a}{c}$.

$$\Rightarrow (1 - \beta_0) e^{kt_0} + (1 + \beta_0) e^{-kt_0} = 2.$$

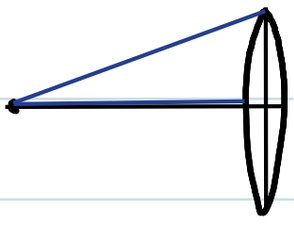
$$\Rightarrow e^{kt_0} = \frac{1 + \beta_0}{1 - \beta_0} \text{ or. } e^{kt_0} = 1.$$

$$\Rightarrow t_0 = \frac{c}{a} \ln \frac{1 + \beta_0}{1 - \beta_0} = \frac{c}{a} \ln \frac{c + v_0}{c - v_0}.$$

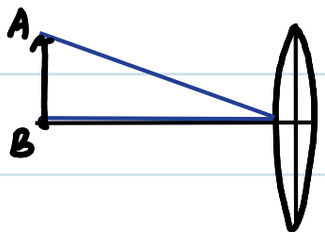
or. $t_0 = 0.$

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单位光：细若.



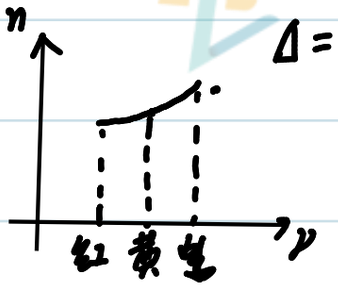
球差.



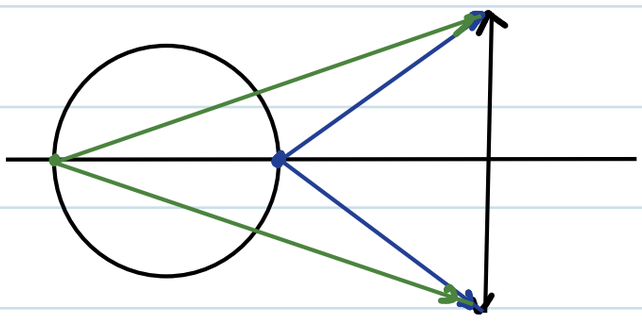
慧差

做题.自动使用傍轴近似.

色差.



$$\Delta = \frac{n_{紫} - n_{红}}{n_{黄} - 1}.$$



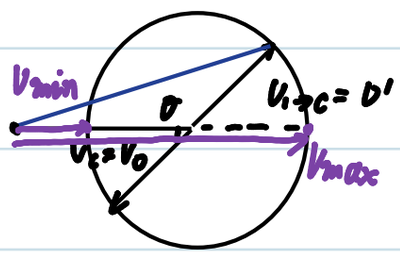
傍轴近似.

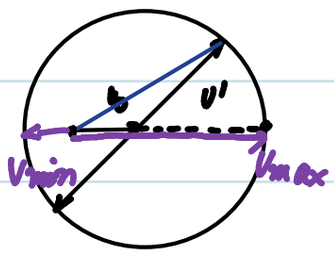
29. 7. 补.

$V_c = ?$

$C: ?$

地：?

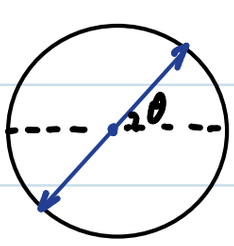




$V_t = V_0$.

质心系 S'. $E' = \frac{1}{2} M_0 C^2$.

$$p'^2 C^2 + m_0^2 C^4 = E'^2.$$



$p_x'^2 + p_y'^2 = p'^2$

$p_x' = p' \cos\theta$.

$p_y' = p' \sin\theta$.

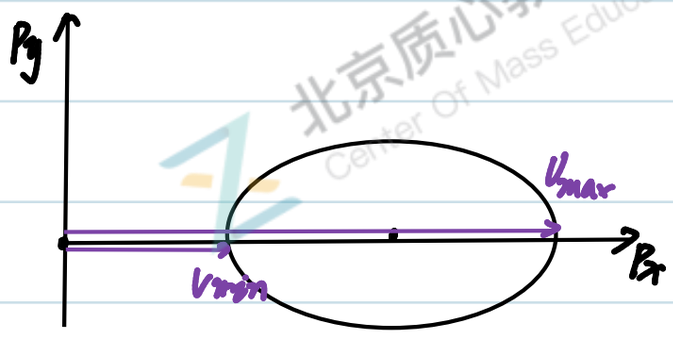
、换到地面系：

$$p_x = \gamma(p_x' + \beta \frac{E'}{c})$$

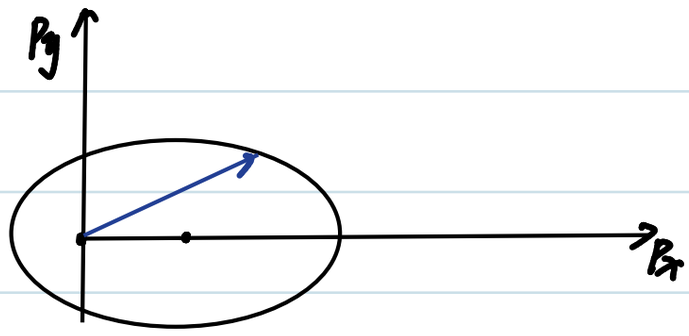
$$p_y = p_y'.$$

$$\Rightarrow p_x = \gamma \cdot p' \cdot \cos\theta + \frac{\beta}{c} E'.$$

$$p_y = p' \sin\theta.$$



$\frac{\beta}{c} E' > \gamma p'$. 则同一维.



在点在焦点外. 同一维.

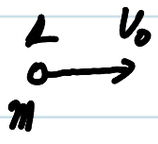
在点在焦点内.

$$p^2 = p_x^2 + p_y^2 = (\gamma \cdot p' \cdot \cos\theta + \frac{\beta}{c} E')^2 + p'^2 (1 - \cos^2\theta).$$

$\frac{dp}{d\theta} = 0 \Longleftrightarrow \frac{dp^2}{d(\cos\theta)} = 0$. 得到最大最小速度.

相对论.补：

非相.



能不接

1球：各自方向飞走.

2球：S水平方向速度最大偏角.

未：V_{1x}. V_{1y} V_{2x}. V_{2y}. V_{3x}. V_{3y}.

方：$\Delta p_x = 0$. $\Delta p_y = 0$. $\Delta E = 0$. 约. $\frac{V_{2y}}{V_{2x}} = \tan\theta$.

$\Rightarrow$ 角在面上.

2.3. 质心. S 2.3的相对.

1. S23质心的质心. 1 S23质心的相对.

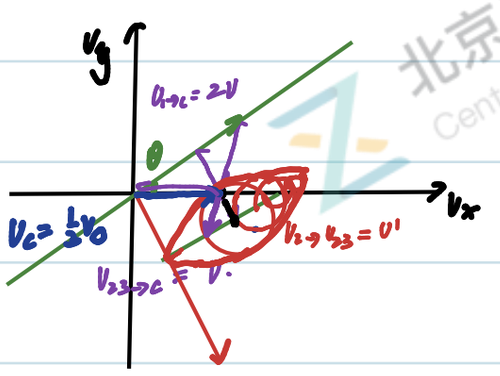
设：$V_{23\to C}=V$. $\Rightarrow V_{1\to C}=2V$.

$V_{2\to 23}=V'$. $\Rightarrow V_{3\to 23}=V'$

能：$\frac{1}{2}mv_0^2=\frac{1}{2}3mv_c^2+\frac{1}{2}\cdot\frac{m\cdot 2m}{3m}(3V)^2+\frac{1}{2}\frac{m\cdot m}{m+m}(2V)^2$.

$\Rightarrow v_0^2=\frac{1}{3}v_0^2+6V^2+2V'^2$.

$\Rightarrow \frac{1}{3}v_0^2=3V^2+V'^2$.



找到椭圆.找到相切.

$2V_{min}=\frac{1}{3}v_0\sin\theta$. $\Leftrightarrow V'_{max}=?$

$V_{max}=?$ $\Leftrightarrow V'_{min}=0$

半短轴曲. $b=V'_{max}$

半长轴曲. a 为 $[\sqrt{V^2-V_{min}^2}+V']$ 的最大值.

相对论：

地面系 S.

$S''\to S'$ 速度为 v.

1.2.3 质心系. S'.

2.3质心系. S'': E_{23}'' 为 S'' 中看到的23总能量.

$\begin{bmatrix} P \\ \frac{E}{c} \end{bmatrix}$ $\begin{bmatrix} 0 \\ \frac{E'}{c} \end{bmatrix}$

$\frac{E}{c}=\gamma(\frac{E'}{c}+\beta\cdot 0)$

$\Rightarrow E=\gamma E'$

↙ ↘ 质心中的总能量

地面看到的总能量.

相对论. 科西第.

质心系：$M_0c^2=\sqrt{(P_{1x}''^2+P_{1y}''^2)c^2+m_0^2c^4}$

$+\gamma_v E_{23}''$ ------- ①

$P_v=-\frac{P_{1x}'\cdot}{E_1'}=-\frac{P_{1x}''\cdot}{\sqrt{(P_{1x}''^2+P_{1y}''^2)c^2+m_0^2c^4}}$

S''：$(P_{2x}''^2+P_{2y}''^2)c^2+m_0^2c^2=\frac{1}{2}E_{23}''$ ----- ②.

$P_{2x}''=P''\cos\theta$

$P_{2y}''=P''\sin\theta$

S'：$P_{1x}=\gamma_0(P_{1x}'+\beta\cdot E_1')$.

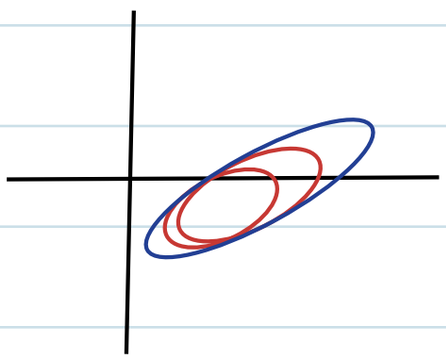
$E_1=\gamma_0(\frac{E_1'}{c}+\beta P_{1x})$.

$P_{1y}=\tan\theta\, P_{1x}$.

换回地面：(洛仑兹).

$P_{2x}=P_{2x}(\cos\theta)$.

$P_{2y}=P_{2y}(\cos\theta)$.



普适数学：曲线簇. $F(x,y,s)=0$.

s 为参.

$\frac{\partial F}{\partial s}=0$. 即得包络线.

变量：P_{1x}. P_{1y}. P_{2x}. P_{2y}. P_{3x}. P_{3y}.

约：$\frac{P_{1y}}{P_{1x}}=\tan\theta$.

和律：$\Delta P_x=0$. $\Delta P_y=0$. $\Delta E=0$.

$P_{1x}+P_{2x}+P_{3x}=\gamma M v_0$

$P_{1y}+P_{2y}+P_{3y}=0$.

$(P_{1x}^2+P_{1y}^2)c^2+(P_{2x}^2+P_{2y}^2)c^2+(P_{3x}^2+P_{3y}^2)c^2+3m_0c^2=\gamma M_0c^2$.

$\Rightarrow F(P_{2x},P_{2y},P_{1x})=0$.

$\frac{\partial F}{\partial P_{1x}}=0$. 得到椭圆形包络线

联立 $P_{2y}=kP_{2x}$. 得. 二次方程. $f(P_{2x})=0$.

$\Delta=0$. 得. 2粒子最大偏转角.

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